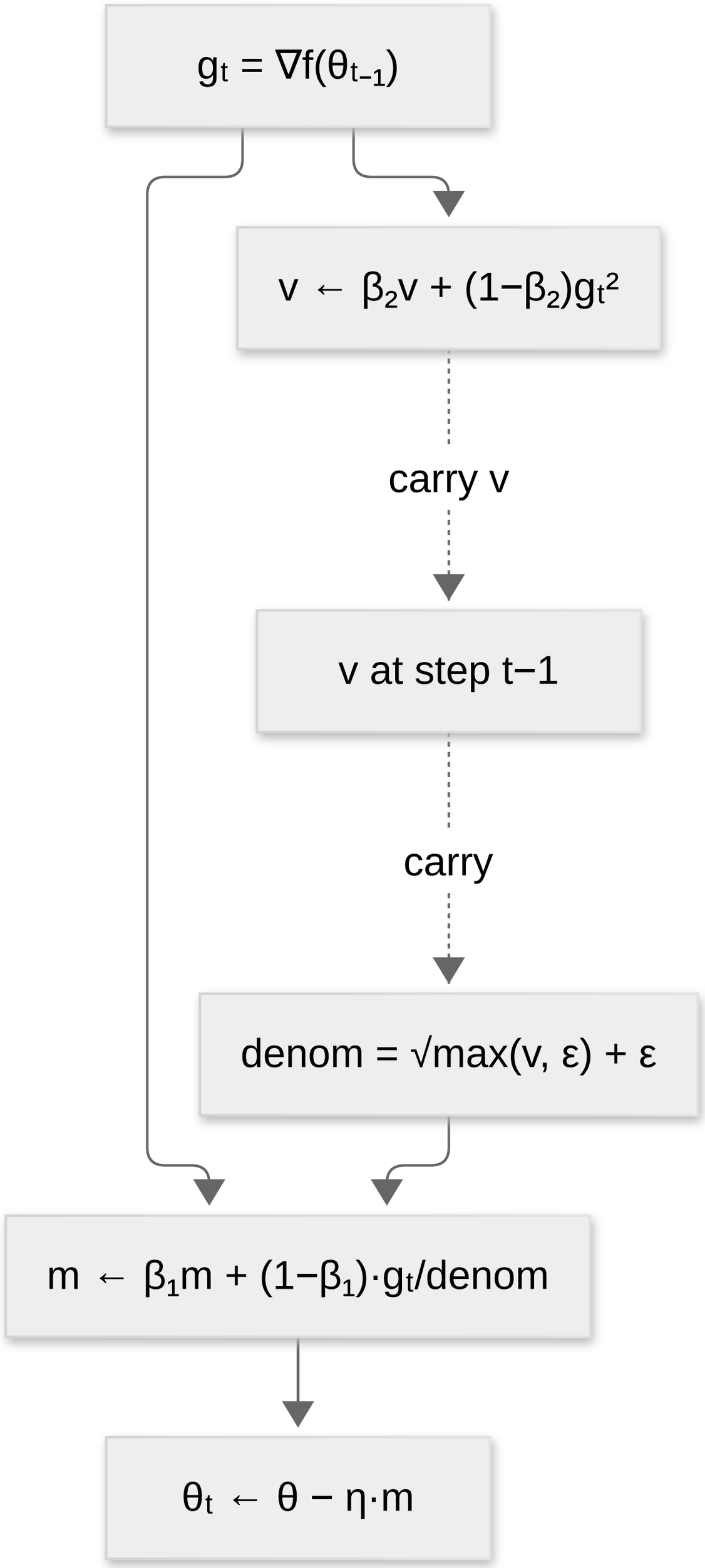

$g_t = \nabla f(\theta_{t-1})$
$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
carry v
v at step t−1
carry
denom = √max(v, ε) + ε
$m \leftarrow \beta_1 m + (1-\beta_1) \cdot g_t / \text{denom}$
$\theta_t \leftarrow \theta - \eta \cdot m$